An annotation-free format for representing multi-modal data features

Tyrone Chen[1], Sonika Tyagi[1,2]

1 School of Biological Sciences, Monash University, Clayton, VIC, 3800, Australia;
2 Department of Infectious Disease, The Alfred Hospital, Melbourne, VIC, 3004, Australia

tyrone.chen@monash.edu 0000-0002-9207-0385

INTRODUCTION

- Integration of various omics layers and health metadata is necessary to understand biological systems holistically for personalised medicine[1-4] (Figure 1a).
- Data formats and standards are diverse, making integration challenging[1-4].

1(a)

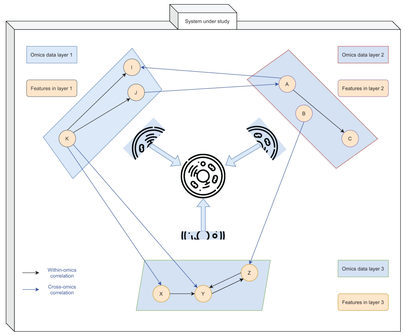
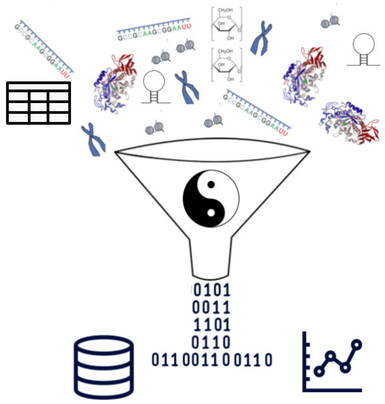


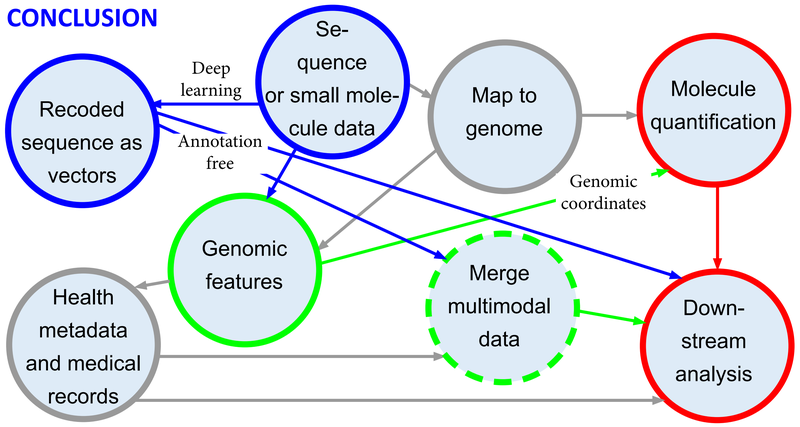
Figure 1a: The rectangles represent different layers of omics data (e.g. proteome, transcriptome and lipidome) while the circles represent features within their respective omics data layer. Black single-line arrows show correlation between features within the omics data (e.g. a regulatory factor) while blue double-lines show correlation between features across different omics data layers.

1(b)



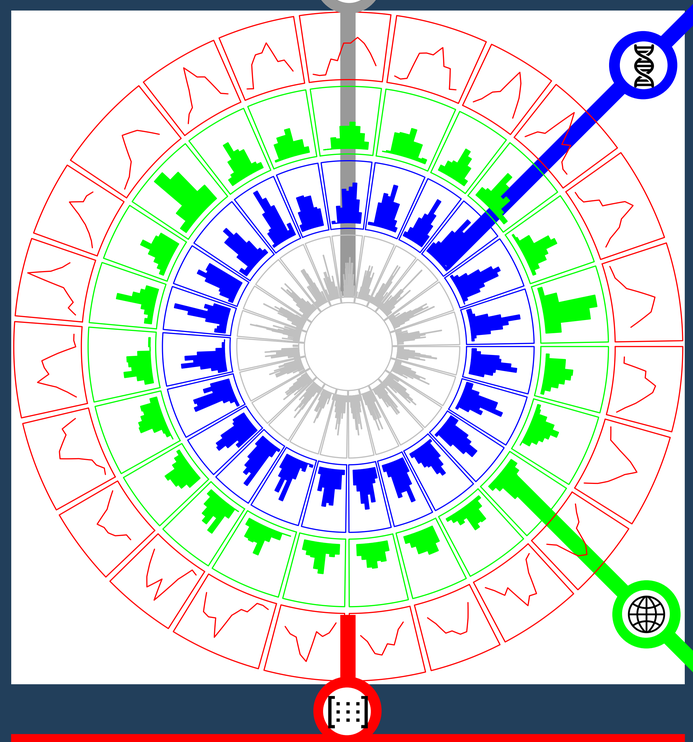
- We propose an intermediate form of data with the aim of unifying high-dimensional and multi-modal data types[2] (Figure 1b).

CONCLUSION



CATEGORICAL DATA: METADATA OR STATIC VARIABLES

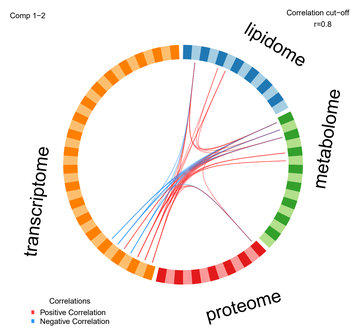
- Categorical variables such as **health metadata** can be integrated with other forms of data[3,4].



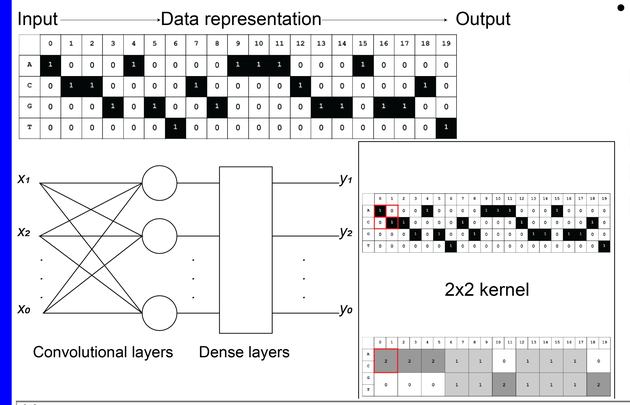
QUANTITATIVE MEASUREMENTS: MATRICES

Figure 3: Using our R "multiomics" pipeline, we integrated multi-omics data[3,4]. The lipidome block is in blue, metabolome block is in green, proteome block is in red and transcriptome block is in orange.

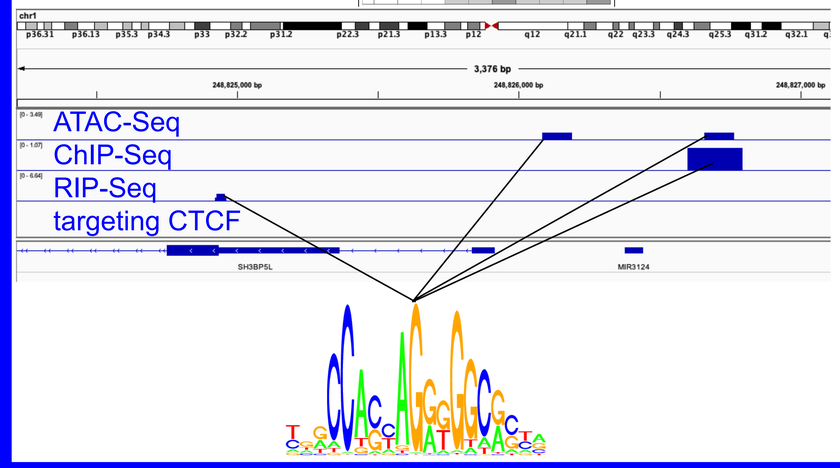
- We use a COVID-19 study with **four omics** data types (lipidomics, metabolomics, proteomics, transcriptomics) and **patient metadata**:



SEQUENCE DATA: FASTA/FASTQ



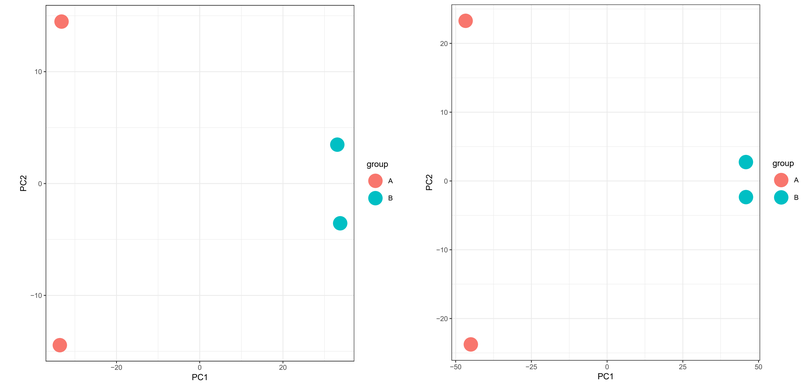
- We demonstrated that **obtaining signal directly from sequence data is possible**, and build upon this deep learning approach[4].



QUANTITATIVE MEASUREMENTS: GENOME COORDINATES

- We reproduce a RNA-Seq analysis result using our **annotation-free format**[2,6,7].

Figure 5: We reproduce an existing single-omics workflow[2,6,7] on an established RNA-Seq dataset[6]. A PCA of samples (a) using a conventional workflow[6,7] and (b) using our data formulation is presented. In both PCAs, sample groups A and B show similar patterns. Both datasets are processed identically, with the exception of our reformatting step in (b). Principal components were computed for filtered CPM data. A script to regenerate this plot is present in our gitlab repository[2].



Our universal multimodal format unifies categorical, sequence and quantitative measurements (matrices, coordinates).

References:
1. Chen T, Tyagi S. Integrative computational epigenomics to build data-driven gene regulation hypotheses, GigaScience, Volume 9, Issue 6, June 2020, giaa064, https://doi.org/10.1093/gigascience/giaa064
2. https://gitlab.com/tyagilab/universal_data_format
3. Chen T, Abadi AJ, Lê Cao KA and Tyagi S. multiomics: A user-friendly multi-omics data harmonisation R pipeline [version 1; peer review: awaiting peer review]. F1000Research 2021, 10:538 (https://doi.org/10.12688/f1000research.53453.1)
4. Chen T, Philip M, Lê Cao KA, Tyagi S. A multi-modal data harmonisation approach for discovery of COVID-19 drug targets, Briefings in Bioinformatics, 2021;, bbab185, https://doi.org/10.1093/bib/bbab185
5. Chen T, Rigby JD, McGee MD and Tyagi S. Deep learning has potential for harmonising multi-omics data to discover weak regulatory features [version 1; not peer reviewed]. F1000Research 2019, 8:2068 (poster) (https://doi.org/10.7490/f1000research.1117749.1)
6. http://bioinf.wehi.edu.au/limma/
7. SEQC/MAQC-III Consortium. A comprehensive assessment of RNA-seq accuracy, reproducibility and information content by the Sequencing Quality Control Consortium. Nat Biotechnol 32, 903–914 (2014). https://doi.org/10.1038/nbt.2957

We acknowledge high performance computing support provided by the Monash eResearch Centre.

We acknowledge and pay respects to the Elders and Traditional Owners of the land on which our four Australian campuses stand.

tyagilab